MB Note

1 HIV|SIR Model

HIV: $\begin{cases}\dot x=\lambda-dx-\beta xv\\ \dot y=\beta xv-ay\\ \dot v=ky-uv\end{cases}\Rightarrow$ λ: formation rate of healthy cells. d: death rate of healthy cells. β: infection rate. a: death rate of infected cells. k: formation rate of free viruses. u: death rate of free viruses.

v 中缺 $-\beta xv$ 的原因: The model neglects the cost of virus when it infects healthy cells.

virus-free steady-state: $y=0$ and $v=0$, possible $x=\frac{\lambda}{d}$.

Extimate virus increase: *in each virus generation|over the characteristic lifetime:*

Have: $(x,y,v)(0)=(\frac{\lambda}{d},0,v_0)$. 1. Period of free virus lifetime: u^{-1}. Period of infected cell lifetime: a^{-1}.

2. $\frac{1}{u}\ll\frac{1}{a}$ 3. $v(\frac{1}{u})-v(0)=\int_0^{\frac{1}{u}}\dot v dt=k\int_0^{\frac{1}{u}}ydt-u\int_0^{\frac{1}{u}}vdt$

a1. $\int_0^{\frac{1}{u}}ydt\simeq\int_0^{\frac{1}{u}}y(\frac{1}{a})dt$ b. $\int_0^{\frac{1}{u}}vdt\simeq\int_0^{\frac{1}{u}}v(0)dt$, a2. $y(\frac{1}{a})=\int_0^{\frac{1}{a}}\dot ydt\simeq\int_0^{\frac{1}{a}}\dot y(0)dt$

Basic Reproductive ratio: $R_0=\frac{k\beta\lambda}{adu}$. $R_0>1$ virus invades; $R_0<1$ virus dies out.

估计 (e.g. R_0) 某个值范围 | 最后趋近于某个值 | 得到趋近速率: 1. 计算 equilibrium. 2. 计算 Jacobian 算 eigenvalues. 3. 取最大 eigenvalue $\lambda\to$ 得 sol $\propto e^{\lambda t}$. 4. 根据题意估计要求值范围 | 趋近值 | 速率 (λ).

(e.g.) 证明 $u=$ cont. when $t=\mathcal{O}(u^{-1})$: 1. Linearise 估算近似值 2. 假设原式 $\simeq$ 估计值 3. 解出 t 估计

SIR: $\begin{cases}S'=-\beta SI\\ I'=\beta SI-\alpha I\\ R'=\alpha I\end{cases}\Rightarrow$ S: susceptible individuals (易感人群). I: infected individuals (感染者). R: recovered individuals (康复者). β: infection rate. α: recovery rate. (or death, if no R)

ps: $\lim_{t\to\infty}I(t)=0$ $\lim_{t\to\infty}S(t)>0$ and $R(0)=0$

Assumption: No births/deaths/migration.
Total Pop.: $N=S+I+R$ (constant).
Basic Reproductive Number: $R_0=\frac{\beta N}{\alpha}$.
$R_0>1\Leftrightarrow S>\frac{\alpha}{\beta}$ (epidemic occurs).
$R_0<1\Leftrightarrow S<\frac{\alpha}{\beta}$ (disease dies out).

Final Size Relation: $\ln(S_0/S_\infty)=\frac{\beta}{\alpha}[N-S_\infty]=R_0(1-\frac{S_\infty}{N})$

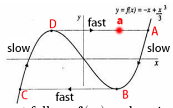
proof: 1. $(S+I)'=-\alpha I$ 2. $\int_0^\infty(S+I)'dt=-\alpha\int_0^\infty Idt\Rightarrow(S+I)_\infty-(S+I)_0=\int_0^\infty\frac{\alpha S'}{\beta S}dt$

3. By $N=S_0+I_0$, can get $S_\infty-N=\ln(S_\infty/S_0)\frac{\alpha}{\beta}$. □

2 Relaxation|Excitation|Switch - FitzHugh Nagumo Model

Explain why oscillations|relaxation occur:

1. (e.g.) $\dot x=c[f(x)-y]$ and $\dot y=$ a line, where $f(x)$ is a cubic-like curve. $c\gg1$
2. show $|\dot x|\gg|\dot y|$ (or vice versa), and no stable equilibrium exists. (can unstable)
3. 图上标一起始点 **a**, 画类似右侧图. 4. (e.g.) a. Start at **a**, $|\dot x|\gg|\dot y|$, so it quickly moves horizontally to the right branch of cubic-like curve (A). b. When it closed to the right branch, since $\dot y<0$, $\dot x\simeq0$, so it moves downwards along the right branch.



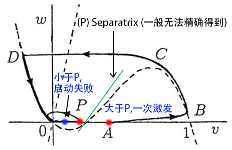
Then, it closes to the point B. It has option to leave the path $f(x)$ since $\dot y<0$, so it cannot follow $f(x)$ and again quickly jumps from B to the left branch of the cubic-like curve (C) since $|\dot x|\gg|\dot y|$. When it closed to the left branch, since $\dot y>0$, $\dot x\simeq0$, so it moves upwards along the left branch. Finally, it reaches the point D, and again quickly jumps to the right branch of the cubic-like curve (A). This process repeats, resulting in relaxation oscillations.

Estimate Period: (neglect fast jumps) 1. Period $T\simeq\int_{t_A}^{t_B}dt+\int_{t_C}^{t_D}dt=2\int_{t_A}^{t_B}dt$ (by symmetry)

2. (e.g.) $\dot x=c[f(x)-y]$, $\dot y=g(x,y)$ then $y\simeq f(x)$, $\frac{dy}{dt}=\frac{dy}{dx}\frac{dx}{dt}=f'(x)\frac{dx}{dt}\Rightarrow dt=\frac{f'(x)}{g(x,y)}dx$

3. Find coordinates of A, B, C, D (turnning points) $\Rightarrow T\simeq2\int_{x_A}^{x_B}\_dx$

Three kinds of Excitable System:
small perturbation: returns to equilibrium. (fail excitation)
large perturbation: large excursion in phase space before returning to equilibrium. (action potential|successful excitation)
Refactory period(不应期): system insensitive to further
constant stimulus: periodic excitation. (repetitive firing|limit cycle)
Explain why excitable occur: 结合前面 relaxation oscillation 的分析, stable equilibrium (图为原点), 分析 P|A 开始 (small|large perturbations).



FitzHugh-Nagumo Model:

Have: $\begin{cases}\dot v=f(v)-w+I(t)\\ \dot w=bv-\gamma w\end{cases}\Rightarrow$ v: membrane (excitable) potential. w: ionic current. (recovery var.) a,b,c: positive constants. I(t): external stimulus.

Switch Behaviour: 类似 relaxation oscillation, 有 two stable equilibria, 分析一个过去后不会回来.

Remark: 对于这类模型, 计算平衡点用 $f(v)$ 的图像斜率和另一条直线斜率来判断.

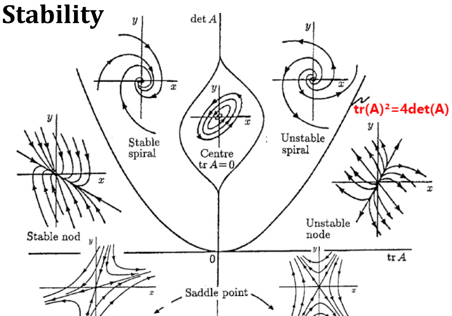
3 Stability|Linearise Stability

Continuous $\frac{dN}{dt}=f(N)$:
- **Stable**: $f'(N^*)<0$
- **Unstable**: $f'(N^*)>0$

Discrete $N_{t+1}=f(N_t)$: $\lambda=f'(N^*)$
- **Stable**: $|\lambda|<1$ - **periodic**: $|\lambda|=1$
- **Unstable**: $|\lambda|>1$ (chaotic)
- **tangent bifurcation**: $\lambda=1$
- **period-doubling bifurcation**: $\lambda=-1$

One var. ↑ | More var. ↓:

Jacobian: $J=[\mathbf{F}_x\ \mathbf{F}_y\ \cdots]$
Eigenvalue Method:
If $\mathrm{Re}(\lambda_i)<0$ for all i, then asy.stable.
If $\lambda_1<0<\lambda_2$, then saddle (unstable).
If $\mathrm{Im}(\lambda_i)\neq0$, then focus (spiral).
If $\mathrm{Re}(\lambda_i)=0$, then center (linear!).
(or(nolinear): **Hopf bifurcation**.



If $\lambda=0$: If $\lambda_1=0$, $\lambda_2<0$: It's a "saddle-node" with one stable eigendirection (corresponding to $\lambda_2<0$) and one "center" eigendirection (corresponding to $\lambda_1=0$). ps: 如果 $\lambda>0$ 或都 $=0$, 类似分析.

Linearise Stability: 1. Let $N(t)=N^*+n(t)$, $|n(t)|\ll1$. or $N_t=N^*+n_t$, $|n_t|\ll1$.

2. $\frac{dn}{dt}=\frac{dN}{dt}=$ a. $\cancel{f(N^*)}^{0}+f'(N^*)n(t)$ or $\cancel{f(N^*)}+\cancel{f'(N^*)}n_t$ b. (delay) 直接代入后 linearise.

Justify bifurcation: 1. Assume $n(t)=e^{\lambda t}$ 2. 代入 $\frac{dn}{dt}$: $\lambda e^{\lambda t}=f'(N^*)e^{\lambda t}$ 3. (Re), (Im) 得两方程.
4a. 反证 $\Re(\lambda)>0,<0$ or **4b.find** T_c Check $T=0$ stable|unstable $\Rightarrow$ first time $\Re(\lambda)=0$

Properties for continuous: **Time-Independent**: At N^*, $N(t-T)=N(t)=N^*$.
Tangency-Condition: At bifurcation point (x^*,r_c), $f_x(x^*,r_c)=f(x^*,r_c)=0$.

Properties for discrete: **eigenvalue**: $\lambda=f'(N^*)$.
(for perturbations) $\lambda<-1$: ↑ oscillation; $-1<\lambda<0$: ↓ oscillation;
$0<\lambda<1$: ↓ monotonic; $\lambda>1$: ↑ monotonic.

p-periodic: $N^*=f^p(N^*)$. Every p-periodic N^* with $\lambda=-1$ branches into a $2p$-periodic N^*.
Sarkovskii's Theorem: If a 3-periodic N^* exists, then p-periodic N^* exists for all $p\in\mathbb{N}^+$.
Max/Min: $f'(N_m)=0$, $N_{\max}=f(N_m)$, $N_{\min}=f(N_{\max})$. (assume unimodal, 2-periodic exists.)

Types of Bifurcations: **Saddle-Node**: $\dot x=r\pm x^2$ (Normal Form) $0\to$ S $\to$ U-S
Transcritical: $\dot x=rx-x^2$ (Normal Form) S-U $\to$ U-S
super-critical Pitchfork: $\dot x=rx-x^3$ (Normal Form) S $\to$ S-U-S
sub-critical Pitchfork: $\dot x=rx+x^3$ (Normal Form) U $\to$ U-S-U
Hopf (平衡点从稳定变为不稳定, 产生 **limit cycle**) **Condition**: $\det(J)>0$, $\frac{d\mathrm{tr}(J)}{d\mu}\neq0$. μ: parameter, not t
or: 找到 μ s.t. $\mathrm{tr}(J)>0$, $\mu>\mu_c$ and $\mathrm{tr}(J)<0$, $\mu<\mu_c$. **Period**: $T=\frac{2\pi}{\omega}$, 特征向量: $\lambda=\pm i\omega$
Bifurcation Diagram: x-axis: equilibrium value; y-axis: parameter value. 虚线: unstable; 实线: stable

 # ODEs

$y'+py=q$: Integrating Factor: $I=e^{\int p\,dt}$ sol: $I(x)y=\int I(x)q\,dt$
$y''-\lambda y=0$: 1. $\lambda>0$: $y=c_1\sinh(\sqrt{\lambda}t)+c_2\cosh(\sqrt{\lambda}t)=c_1'e^{\sqrt{\lambda}t}+c_2'e^{-\sqrt{\lambda}t}$
2. $\lambda=0$: $y=c_1t+c_2$ 3. $\lambda<0$: $y=c_1\sin(\sqrt{-\lambda}t)+c_2\cos(\sqrt{-\lambda}t)$
$ay''+by'+cy=0$: 1. $\Delta>0$: $y=c_1e^{r_1t}+c_2e^{r_2t}$ $r_{1,2}=\mu+vi$
2. $\Delta=0$: $y=e^{rt}(c_1+c_2t)$ 3. $\Delta<0$: $y=e^{\mu t}(c_1\cos vt+c_2\sin vt)$

4 Nullclines|Limit Cycle|Trapping Region

Nullclines: For: $u'=f(u,v)$, $v'=g(u,v)$. u-nullcline: $f(u,v)=0$. v-nullcline: $g(u,v)=0$.
Slope of Nullclines: Since $df=f_udu+f_vdv=0$ on u-nullcline,
Slope of u-nullcline: $\frac{dv}{du}=-\frac{f_u}{f_v}$. Similarly, $\frac{dv}{du}=-\frac{g_u}{g_v}$ for v-nullcline.
Trapping Region: 1. 直线/nullcline 围闭合区域. 2. 边界上, 外法向量 $\mathbf{n}$ 有 $\mathbf{n}\cdot\begin{bmatrix}f(u,v)\\g(u,v)\end{bmatrix}<0$
Poincaré-Bendixson: One unstable equilibria (no saddle) + trapping region = stable limit cycle exists.
Useful Tools Polar Coor.: $rr'=xx'+yy'$ and $r^2\theta'=xy'-yx'$. $\Delta u=u_{rr}+\frac{1}{r}u_r+\frac{1}{r^2}u_{\theta\theta}$

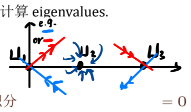
5 Diffusion|Wave Model

Diffusion Eq.: $u(\mathbf{x},t)$ s.t. $u_t=D\Delta u$ with $u(\mathbf{x},0)=f(\mathbf{x})$ D: diffusion coeff.

Fundamental sol: $\phi(\mathbf{x},t)=\frac{Q}{(4\pi Dt)^{n/2}}e^{-\frac{|\mathbf{x}|^2}{4Dt}}$ · If $f(\mathbf{x})=Q\delta(\mathbf{x})$, $u(\mathbf{x},t)=\phi(\mathbf{x},t)$
General sol: $u(\mathbf{x},t)=\int_{\mathbb{R}^n}\phi(\mathbf{x}-\mathbf{y},t)f(\mathbf{y})d\mathbf{y}$ ↓ $v(\mathbf{x},0)=f(\mathbf{x})$
Non-Homo: If $u_t\to u_t=D\Delta u+a(t)u$, consider: $u(\mathbf{x},t)=v(\mathbf{x},t)e^{\int_0^ta(s)ds}$ with $v_t=D\Delta v$.
Flux (J): 1. 考虑 u, J 为 continuous 函数. (尤其是分段函数) 2. J 考虑物理意义, 边界上为 0.
e.g. $J=-\mu b_x+\delta b(\ln s)_x$ μ: diffusion coeff. $\frac{\delta}{s}$: chemotactic sensitivity
Interpret Initial val. $u_0\delta$: Assumption that the { species } were released at a single point and nowhere else.
Give initial condition: Usually using $u(\mathbf{x},0)=u_0$ (uniform distribution)
Give boundary condition: Using u, J is continuous at boundary. $\Rightarrow$ 考虑边缘要求连续得到条件
Wave Model: Consider $u(x,t)=u(z)=U(z)$ with $z=x-ct$ c: wave speed
Combaine diffusion: $u_t=Du_{xx}+f(u)$ $\Rightarrow$ ODE: (换元后得到) $-cU'=DU''+f(U)$

ODEs: (Let $V=U'$) $\begin{cases}U'=V\\ V'=-\frac{c}{D}V-\frac{f(U)}{D}\end{cases}\Rightarrow$ If heteroclinic: If wavefront: (or) $\begin{cases}U(\pm\infty)=u^\pm\\ (u^\pm,0)\text{ are equilibria of ODEs}\end{cases}$

Homoclinic: $U(\pm\infty)=u^*$ (i.e. $u^+=u^-$) **ahead of wave**: u^- **behind wave**: u^+
Condition if wavefront: $\lim_{z\to\pm\infty}U(z)=u^\pm$ and $\lim_{z\to\pm\infty}U'(z)=0$
Show the existence of wavefront|connection|heteroclinic between u^- and u^+:
1. 确定 $u^\pm$, 题目会给, or 由 $f(u)$ 的零点确定. 2. 计算 Jacobian, 确定 $u^\pm$ 的类型且计算 eigenvalues.
3. (ps: for $J=[0,1;a,b]$ 有 eigenvectors: $[1;\lambda]^\top$) The positive|negative eigenvalues at $u^-|u^+$ corresponding to an eigenvector $[1;\lambda]^\top$, which means that it has a unstable|stable eigendirection with positive|negative slope. 4. Therefore, it's plausible



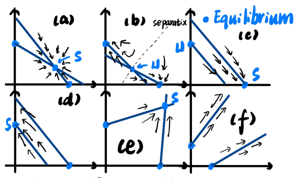
Wave Speed: 1. For $-cU'=DU''+f(U)$ 两边乘 U' 对 z 在 $(-\infty,\infty)$ 积分 $=0$
2. $-c\int_{-\infty}^{\infty}(U')^2dz=D\int_{-\infty}^{\infty}U''U'dz+\int_{-\infty}^{\infty}f(U)U'dz$ by $U'dz=dU$, $\int_{-\infty}^{\infty}U''U'dz=\frac{1}{2}[(U')^2]_{-\infty}^{\infty}$
3. $c=[\int_{u^-}^{u^+}f(U)dU]/[\int_{-\infty}^{\infty}(U')^2dz]$ Remark: Sign of c depends on *area* between $f(U)$ and U-axis
Meaning: critical front speed: $c=0$ (对于灾害蔓延): $c<0|c>0$: no spread(dies out)| spread(occurs)
Possible $c=0$: Assume *continuous* dependence of c on {parameter p}. 证明有 $c>0|c<0\to\exists p_c$ s.t. $c=0$
Fisher-KPP: $u_t=Du_{xx}+ru(1-\frac{u}{K})\Leftrightarrow-cU'=DU''+ru(1-\frac{u}{K})$, $\exists$ wavefront s.t. $c\geq c_{\min}=2\sqrt{rD}$.
Condition for Wavefront sol: eigenvalue|seperation by var. 算出来后 NO oscillation sol. (i.e. NO sin / cos)
Remark: 如遇计算 parameter 值 | 计算待定系数 | 算 wave speed sign: 直接积分或乘 g, U 分部积分!

6 Lotka-Volterra Model —Predator-Prey Model

Model: $\begin{cases}\dot x=x(a-by)\\ \dot y=y(cx-d)\end{cases}\Rightarrow$ x: prey pop. a: prey growth rate. b: predation rate coeff. y: predator pop. c: predator growth rate coeff. d: predator death rate.

Constant of Motion: 1. 用 $\frac{dy}{dx}=\frac{\dot y}{\dot x}$ 算出 trjectory 方程 e.g. $H(u,v)=\alpha u+v-\ln(u^\alpha v)=H_0$
2. If $\frac{dH}{dt}=0$ or $\frac{dH}{d\tau}=0$, H is **constant of motion**.
Show trajectory closed: a. Calculate $\begin{bmatrix}H_{uu}&H_{uv}\\H_{vu}&H_{vv}\end{bmatrix}$ at equilibrium (u^*,v^*).
b. If positive|negative definite ($\lambda_i>0|\lambda_i<0$), then H strictly convex|concave, has global min|max, closed.
c. Therefore, trajectory is closed. **Closed trajectory** $\Rightarrow$ **limit cycle.** (periodic sol)
Why periodic occurs: As any trajectory must lies on one of the closed level curves of H, and we have shown that it is closed. Thus, the periodic oscillation occurs.

Principal of Competitive Exclusion:
If two species compete for the same limited resource, one will outcompete and exclude the other. ps: 图 b,c,d 都是.
Competition Model:
$\begin{cases}\dot x=x(r_1-b_1x-c_1y)\\ \dot y=y(r_2-b_2y-c_2x)\end{cases}\Rightarrow$ x, y: pop. of species 1,2; b_i: intraspecific competition; c_i: interspecific competition



右图描述: **(a)**: Stable coexistence. (Weak competition)
intraspecific competition > interspecific competition
(b): Bistability. (Strong competition, 由 initial condition 决定)
interspecific competition > intraspecific competition
(c), (d): One species outcompetes the other. interspecific competition > intraspecific competition
(e): Weak mutualism. Both species tend to a stable coexistence, benefiting from each other
(f): Strong mutualism. Due to absence of attractor, populations grow without bound, resulting population explosion.

7 Reaction | Reduced, Layer Problem

Rate of uptake: $v=\frac{d[P]}{dt}|_{t=0}$
Complex: 对于反应 e.g. $S+E\overset{b}{\underset{a}{\rightleftharpoons}}SE\overset{c}{\to}P+E$. E: enzyme (酶), SE: complex. 快速出现后稳定, P: 产物
Law of Mass Action: e.g. For reaction $\alpha_1A+\alpha_2B\overset{b}{\underset{a}{\rightleftharpoons}}\beta_1C+\beta_2D$ e.g. $\frac{d[C]}{dt}=b[A]^{\alpha_1}[B]^{\alpha_2}-a[C]^{\beta_1}[D]^{\beta_2}$
Singular Perturbation Systems: $0<\varepsilon\ll1$

fast $\begin{cases}\dot u=f(u,v,\varepsilon)\\ \dot v=\varepsilon g(u,v,\varepsilon)\end{cases}\overset{\tau=\varepsilon t}{\Longleftrightarrow}$ slow $\begin{cases}\varepsilon u'=f(u,v,\varepsilon)\\ v'=g(u,v,\varepsilon)\end{cases}\overset{\varepsilon=0}{\Longrightarrow}$ layer $\begin{cases}\dot u=f(u,v,0)\\ \dot v=0\end{cases}$ reduced $\begin{cases}0=f(u,v,0)\\ v'=g(u,v,0)\end{cases}$

Inner sol: sol of fast time (layer prob.) **Outer sol:** sol of slow time (reduced prob.) | t: fast var., v: slow var.
Quasi-Steady State Approximation (QSSA): Assume slow system $\varepsilon\to0$ (reduced), find outer sol.
Solve inner|outer sol — reduced|layer prob.: (还可直接计算, 对于 manifold 类型的)

1.Assume: Inner sol $\begin{cases}U(t;\varepsilon)=u(\tau;\varepsilon)=\sum_{n=0}^\infty\varepsilon^nU_n(t)\\ V(t;\varepsilon)=v(\tau;\varepsilon)=\sum_{n=0}^\infty\varepsilon^nV_n(t)\end{cases}\overset{\tau=\varepsilon t}{\Longleftrightarrow}$ Outer sol $\begin{cases}u(\tau,\varepsilon)=\sum_{n=0}^\infty\varepsilon^nu_n(\tau)\\ v(\tau,\varepsilon)=\sum_{n=0}^\infty\varepsilon^nv_n(\tau)\end{cases}$

2. Substitute into layer|reduced prob., compare coeff. $\mathcal{O}(1)$ (i.e. $\varepsilon=0$)
3. Initial condition: For **inner**: $U_0(0)=u(0)$, $V_0(0)=v(0)$; but $v_0(0)\neq u(0)$.
For **outer**: (valid) $v_0(0)=v(0)$, $u_0(\tau)=v_0(\tau)$, $u_0(0)=v_0(0)$; (unvalid; mismatch) also: $u_0(0)=u(0)$, ↑
4. **Remark:** 先写 outer, 再写 inner. 为啥写 inner 原因: we also have $u_0(0)=u(0)$, but $u_0(0)=v_0(0)\neq u(0)$, it's inconsistent with the initial condition of the outer solution. Thus we need to consider the boundary layer. We using $t=\frac{\tau}{\varepsilon}$ to scale the fast time so that we can capture the rapid change in u near $\tau=0$.
Asymptotic Matching: $\lim_{t\to\infty}[U_0(t),V_0(t)]=\lim_{\tau\to0^+}[u_0(\tau),v_0(\tau)]$
Activator-Inhibitor System:
$\begin{cases}\dot u=f(u,v)\\ \dot v=g(u,v)\end{cases}$ If $g_u>0$ (<0): u is **activator** (**inhibitor**) of v; If $f_v<0$ (>0): v is **inhibitor** (**activator**) of u

8 logistic Model | Linearise

Model: $\frac{dN}{dt}=rN(1-\frac{N}{K})-p(N)$, r: birth/growth rate. K: carrying capacity. $p(N)$: predation term (捕食者).

1. Binomial: $(1+x)^a\simeq1+ax$ for $|x|\ll1$ Exp: $e^{ax}\simeq1+ax+\frac{(ax)^2}{2!}$ for $|x|\ll1$
2. **Linearise**: $f(x+\Delta x)\simeq f(x)+f'(x)\Delta x$ for $|\Delta x|\ll1$ (一般用这个没问题)
3. ln: $\ln(x+\Delta x)\simeq\ln(x)+\frac{1}{x}\Delta x$ for $|\Delta x|\ll|x|$ Geometric Series: $\frac{1}{1-x}\simeq1+x+x^2$ for $|x|\ll1$
4. **Integral Approx.**: $y(a)-y(b)=\int_b^ay'dx\simeq(a-b)y'(a)\simeq(a-b)y'(b)$
5. **Taylor**: $f(x,y)\simeq f(x_0,y_0)+f_x(x_0,y_0)(x-x_0)+f_y(x_0,y_0)(y-y_0)$ for $|x,y-x_0,y_0|\ll1$

$x'=a+bx$: Sol: $x(t)=-\frac{a}{b}+ce^{bt}$ $dA=rdrd\theta$ $dV=r^2\sin\phi\,drd\theta d\phi$

 # 条件表达

Radial Symmetry: $u(r,\theta)$ s.t. $u_\theta=0$ **Equilibrium:** $u(\mathbf{x},t)$ s.t. $u_t=0$
Stationary|Steady-State|Static: $u(\mathbf{x},t)$ s.t. $u_t=0$ **Spatially Homo:** $u(\mathbf{x},t)$ s.t. $u_\mathbf{x}=0$
Homo Rest State|Const. Equilibrium|Uniform Steady State: $u_\mathbf{x}=0$ and $u_t=0$

9 Manifolds|Fenichel's Theorem

(Recall) Singular Perturbation Systems: $0 < \varepsilon \ll 1$

fast $\begin{cases} \dot u = f(u, v, \varepsilon) \\ \dot v = \varepsilon g(u, v, \varepsilon) \end{cases} \overset{\tau = \varepsilon t}{\Longleftrightarrow}$ **slow** $\begin{cases} \varepsilon u' = f(u, v, \varepsilon) \\ v' = g(u, v, \varepsilon) \end{cases} \overset{\varepsilon = 0}{\rightarrow}$ **layer** $\begin{cases} \dot u = f(u, v, 0) \\ \dot v = 0 \end{cases}$ **reduced** $\begin{cases} 0 = f(u, v, 0) \\ v' = g(u, v, 0) \end{cases}$

ps: fast var. u (can be *vector*); slow var. v.

Critical Manifold: $M_0 := \{(u, v) : f(u, v, 0) = 0\}$ ps: reduced problem is defined on M_0

Normal Hyperbolicity: Critical manifold M_0 is normally hyperbolic if:

[$\Re(\lambda_i) \neq 0$ (特征值) of $J = [f_{\mathbf{u}_1}\ f_{\mathbf{u}_2}\ \cdots]$] $\Rightarrow$ For u 1-dim: If $\frac{\partial f}{\partial u} \neq 0$

If $\Re(\lambda_i) < 0\ (> 0) \Rightarrow \frac{\partial f}{\partial u} < 0\ (> 0)$: M_0 is **normally attracting** (stable) (**normallyrepelling** (unstable)).

Fenichel's Theorem: If $M_0 \subseteq \{(u, v) : f(u, v, 0) = 0\}$ is compact (closed & bounded) and normally hyperbolic. Suppose f, g are smooth. Then for $\varepsilon > 0$ (sufficiently small), $\exists$ Manifold M_ε s.t. it's $\mathcal{O}(\varepsilon)$ close to M_0 and diffeomorphic to M_0. (i.e. M_ε is locally invariant under the flow of the full system.) M_ε called **slow manifold**.

Find Slow Manifold M_ε: If $M_0 = \{(u, v) : u = h_0(v)\}$, then $M_\varepsilon = \{(u, v) : u = h_\varepsilon(v)$

e.g. Given exact sol: $(u, v) = ((u_0 - \frac{v_0^2}{1-2\varepsilon})e^{-t} + \frac{v_0^2}{1-2\varepsilon} e^{-2\varepsilon t}, v_0 e^{-\varepsilon t})$. Given $M_0 = \{(u, v) : u = v^2\}$.

Then **1.** Linearise, $\tau = \varepsilon t \Rightarrow (u, v) = (\frac{v_0^2}{1-2\varepsilon} e^{-2\tau}, v_0 e^{-\tau}) \Rightarrow u = \frac{v^2}{1-2\varepsilon}$. **2.** Check, when $\varepsilon = 0$, $u = v^2$.

3. Therefore, $M_\varepsilon = \{(u, v) : u = \frac{v^2}{1-2\varepsilon}\}$. If $[u]' = [\frac{v^2}{1-2\varepsilon}]'$, it's **invariant**. (under slow flow)

Invariance of Manifold: $M_\varepsilon = \{(u, v) : u = h_\varepsilon(v)\}$, if $[u]' = [h_\varepsilon(v)]'$, $' = d/d\tau$ 或对应的.

10 Havesting Model

Discrete Population Model: $N_{t+1} = N_t + rN_t(1 - \frac{N_t}{K}) - H_t$, ps: 分析类似 continuous model.

- If $N_{\min} < 1$ **species eradicated/extinct.**
- **Recovery Time**: $T = -\frac{1}{\lambda}$, where $\lambda = f'(N^*)$

Continuous Population Model: $\frac{dN}{dt} = rN(1 - \frac{N}{K}) - h(N)$, r: growth rate, K: carrying capacity.

$h(N)$: harvesting term. a.constant: $h(N) = Y_0$; b.density-dependent: $h(N) = EN$ (E: harvesting effort).

For the density-dependent harvesting, we have:

- If $E > r$, species eradicated/extinct. If $E < r$, we have the following:
- **Maximum (Sustainable) Yield**: $= \max_N h(N) = h(\frac{K}{2}) = \frac{rK}{4}$ ps: at $N^* = \frac{K}{2}$, $E = \frac{r}{2}$

Improved our Model: Combine both harvesting types: $h(N) = EN$ for N small, $h(N) = Y_0$ for N large.

coalesce: 两点合并 eradicated: 根除 membrane potential: 膜电位 unimodal: 单峰 vaccination: 疫苗接种
predator: 捕食者 prey: 被捕食者 trnscriptase: 反转录酶 invades: 入侵 plausible: 似乎合理的
invasion: 入侵 intraspecific competition: 种内竞争 interspecific competition: 种间竞争
promote: 促进 suppress: 抑制 eliminate: 消除 repelling: 排斥的
boundary region = trapping region = invariant region = confined region

Taylor: $\sin(x) \simeq x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$ $\cos(x) \simeq 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$ $\tan(x) \simeq x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots$

$\sinh(x) \simeq x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots$ $\cosh(x) \simeq 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$ $\tanh(x) \simeq x - \frac{x^3}{3} + \frac{2x^5}{15} - \cdots$

Hyperbolic: $\cosh^2(x) - \sinh^2(x) = 1$ $1 + \tanh^2(x) = \mathrm{sec}\,h^2(x)$ $\sinh(x) = \frac{e^x - e^{-x}}{2}$ $\cosh(x) = \frac{e^x + e^{-x}}{2}$

Integral: $\int \frac{1}{\sqrt{a^2 - x^2}} dx = \arcsin(\frac{x}{a})$ $\int \frac{1}{\sqrt{x^2 - a^2}} dx = \mathrm{arcosh}(\frac{x}{a})$ $\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \arctan(\frac{x}{a})$

$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \mathrm{arcsinh}(\frac{x}{a})$ $\int \frac{1}{x^2 - a^2} dx = \frac{1}{a} \mathrm{arctanh}(\frac{x}{a})$

11 Regular/Singular Perturbation Equation

Singular: If eq like: $\varepsilon y'' + a(x, \varepsilon)y' + b(x, \varepsilon)y = 0$ (i.e. highest der. mult. by ε)

Regular: If eq like: $y'' + a(x, \varepsilon)\varepsilon y' + b(x, \varepsilon)y = 0$ (i.e. highest der. NOT mult. by ε)

Approximation Solution: Assume sol: $y(x; \varepsilon) = y_0(x) + \varepsilon y_1(x) + \varepsilon^2 y_2(x) + \cdots$

Substitute into ODE, compare coeff. of $\mathcal{O}(1), \mathcal{O}(\varepsilon), \mathcal{O}(\varepsilon^2)$, etc. 不留 $Y_0 = 0$ 解, 前面已经得到过 $\downarrow$

- If **singular**, we cannot approximate the other root. $\rightarrow Y = \varepsilon y$, and get new ODE.
- Assume sol: $Y(x; \varepsilon) = Y_0(x) + \varepsilon Y_1(x) + \varepsilon^2 Y_2(x) + \cdots$, substitute, compare $\mathcal{O}(1), \mathcal{O}(\varepsilon), \mathcal{O}(\varepsilon^2)$, etc.

***Uniformly Valid Approximation/Expression:** sol = outer sol + inner sol - matching part

where matching part = section.7 Asymptotic Matching 的结果